2-FIBRE PRODUCTS OF CATEGORIES FIBRED IN GROUPOIDS ARE FIBRED IN GROUPOIDS

THE STACKS PROJECT AUTHORS

Editorial note. Counted span is the author's own proof environment, source lines 7453-7463, following the statement at 7445-7451.

Editorial note. The groupoid step (lines 7461-7462) is a componentwise-inverse check against the displayed construction of the cited lemma, recorded as a routine obligation in the card.

Allowed standard prerequisites (editorial).

- 2-fibre products of fibred categories are fibred. lemma-2-product-fibred-categories-over-C, cited at proof lines 7454-7456. Supplies that the explicit 2-fibre product construction is a fibred category over C.
- Fibres of the 2-fibre product over C. lemma-fibre-2-fibre-product-categories-over-C, cited at proof lines 7459-7460. Supplies the identification of the fibre of the 2-fibre product with the 2-fibre product of the fibres, reducing the claim to groupoids.
- Explicit 2-fibre product of categories. lemma-2-fibre-product-categories, cited at proof lines 7461-7462. Supplies the triple construction against which the groupoid property is a componentwise-inverse check.

Source reference. Original lemma 02X9: sources/stacks-categories/pinned-original-source.tex, source lines 5830–5830. Navigation pointer only: the counted proof cites this statement by \ref as a standing input; the deduction the counted proof has to make is carried out inside the counted span, and no part of that deduction is delegated to this pointer.

Source reference. Original lemma 0040: sources/stacks-categories/pinned-original-source.tex, source lines 6298–6298. Navigation pointer only: the counted proof cites this statement by \ref as a standing input; the deduction the counted proof has to make is carried out inside the counted span, and no part of that deduction is delegated to this pointer.

Source reference. Original lemma 02XQ: sources/stacks-categories/pinned-original-source.tex, source lines 6775–6775. Navigation pointer only: the counted proof cites this statement by \ref as a standing input; the deduction the counted proof has to make is carried out inside the counted span, and no part of that deduction is delegated to this pointer.

Source reference. Original lemma 02XI: sources/stacks-categories/pinned-original-source.tex, source lines 6370–6370. Navigation pointer only: the counted proof cites this statement by \ref as a standing input; the deduction the counted proof has

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to make is carried out inside the counted span, and no part of that deduction is delegated to this pointer.

Source reference. Original lemma 003V: sources/stacks-categories/pinned-original-source.tex, source lines 7253–7253. Navigation pointer only: the counted proof cites this statement by \ref as a standing input; the deduction the counted proof has to make is carried out inside the counted span, and no part of that deduction is delegated to this pointer.

Lemma 0.1. *Let $\mathcal{C}$ be a category. The 2-category of categories fibred in groupoids over $\mathcal{C}$ has 2-fibre products, and they are described as in Lemma 0040.*

Proof. By Lemma 02XQ the fibre product as described in Lemma 0040 is a fibred category. Hence it suffices to prove that the fibre categories are groupoids, see Lemma 003V. By Lemma 02XI it is enough to show that the 2-fibre product of groupoids is a groupoid, which is clear (from the construction in Lemma 02X9 for example). □